

# Measuring the footprint of large language models in the astronomy literature

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## ABSTRACT

Large language models became easy to use in late 2022, and their use in scientific writing has grown quickly in most fields, but the size of this change in astronomy has not been reported. We measure it from 200,547 astro-ph abstracts posted to arXiv between 2015 and 2026. The fraction of abstracts that contain a “marker” word, rare before ChatGPT and common after, rose from 1.5% before 2022 to 5.8% in 2025, while a matched set of neutral words stayed flat. A conservative lower bound puts the model-touched fraction of 2025 abstracts near 4%, robust to baseline and word-list choice. Chosen without an imported list, the data recover the same style words next to the genuinely new instrument names, and the marker words occur together far more often than chance (27 of 91 pairs above independence at  $p < 0.01$ ). Against these numbers, explicit disclosure of model use stays below 1% and crude source-code checks find near-zero pasted text. All 22,547 cited arXiv identifiers in 2,532 recent papers resolve, so fabricated citations, which are rising in biomedicine, sit below a 0.24% per-paper limit here. The word signal weakens in 2026 even as disclosure rises. All data and code are public.

*Keywords:* Astronomy data analysis — Scholarly communication — Natural language processing

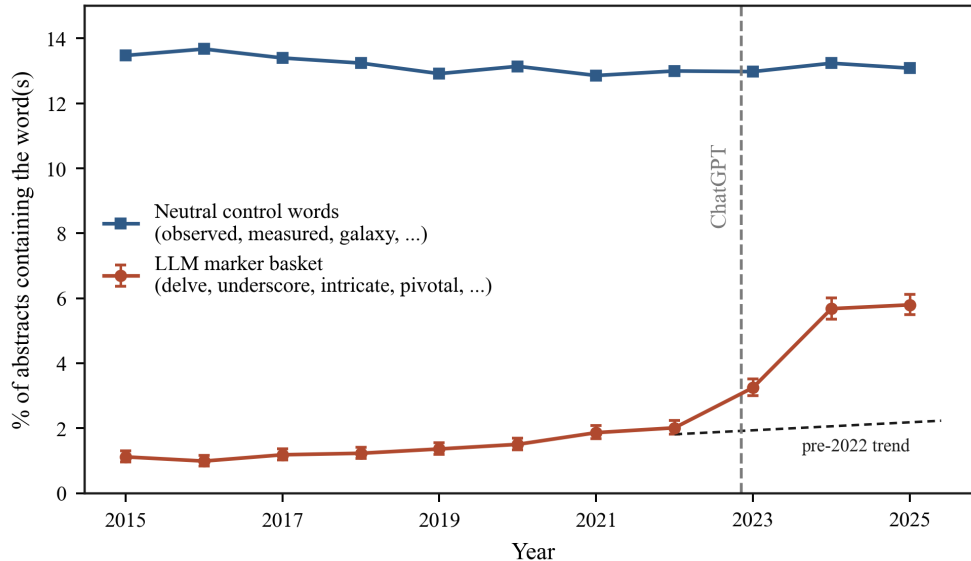
## 1. INTRODUCTION

The public release of ChatGPT in November 2022 made large language models easy to use, and their use in scientific writing followed quickly. Because per-document detectors are not reliable, recent work measures the effect at the level of a whole corpus, by tracking words whose frequency changed abruptly after 2022. [Liang et al. \(2024\)](#) fit a statistical model to arXiv, bioRxiv, and the Nature journals and report that up to about 17.5% of computer science abstracts by early 2024 carried model-edited text. [Kobak et al. \(2025\)](#) count “excess” vocabulary in more than 15 million PubMed abstracts and find that at least 13.5% of 2024 abstracts were processed by a model. [Gray \(2025\)](#) estimate that a model was involved in more than 10% of all 2024 papers and describe a cat-and-mouse effect in which words lose their signal once they become known. Physics and mathematics sit lower in these studies, near 6–10%.

Astronomy has been studied less. [Astarita et al. \(2024\)](#) count model-favored words across about one million records in the NASA Astrophysics Data System ([Kurtz et al. 2000](#)) and show a clear rise after 2023, larger in non-refereed work; they do not convert the signal into a fraction. [Geng & Trotta \(2025\)](#) report the cat-and-mouse effect across all of arXiv. A survey of astronomers ([Fouesneau et al. 2024](#)) finds that most use these tools every week but few state so in print. No paper has reported, for astronomy specifically, how large the footprint is, how it compares with other fields, or how the different ways of detecting it agree.

The size of the effect matters for practical reasons. If a large share of abstracts are edited by a model, then word-based screening of papers or proposals will flag honest authors, and in particular non-native English speakers, at high rates. If model use is common but rarely stated, a reader cannot tell how a given text was produced. And if models draft text rather than only polish it, the record may drift toward a narrower and more uniform style. It is also useful to separate three uses that are easy to run together: a model as a copy editor, a model that drafts content, and a paper produced whole. The methods here mostly measure the first, and a measurement of its size is a first step toward any of these questions.

We provide those numbers. We harvest every astro-ph abstract from 2015 to 2026 and measure the marker-word signal, its rise, and its recent weakening. We convert the signal into a lower bound on the fraction of abstracts a model touched. We let the data select the words and two-word phrases without an imported list, and we measure how often the marker words occur together. We then compare the word signal against three other lenses on the same corpus: explicit disclosure in acknowledgments, pasted-model text in the LaTeX source, and fabricated citations. We close



**Figure 1.** Fraction of astro-ph abstracts containing a marker word (red) and a matched set of neutral control words (blue), by year. Error bars on the marker curve are Wilson 95% intervals; most are comparable to the symbol size ( $n \simeq 15,000$ – $22,000$  abstracts per year). The black dashed segment extrapolates the 2015–2021 linear trend; the excess over it after 2022 is the signal. Both curves are per-abstract fractions, so we stop at the last full year, 2025.

with cuts by subfield and by author country, a measurement of promotional versus hedging language, and the limits of the method. The point is narrow and worth stating up front: a marker word is a sign of model-edited language, not of poor or dishonest science.

## 2. DATA

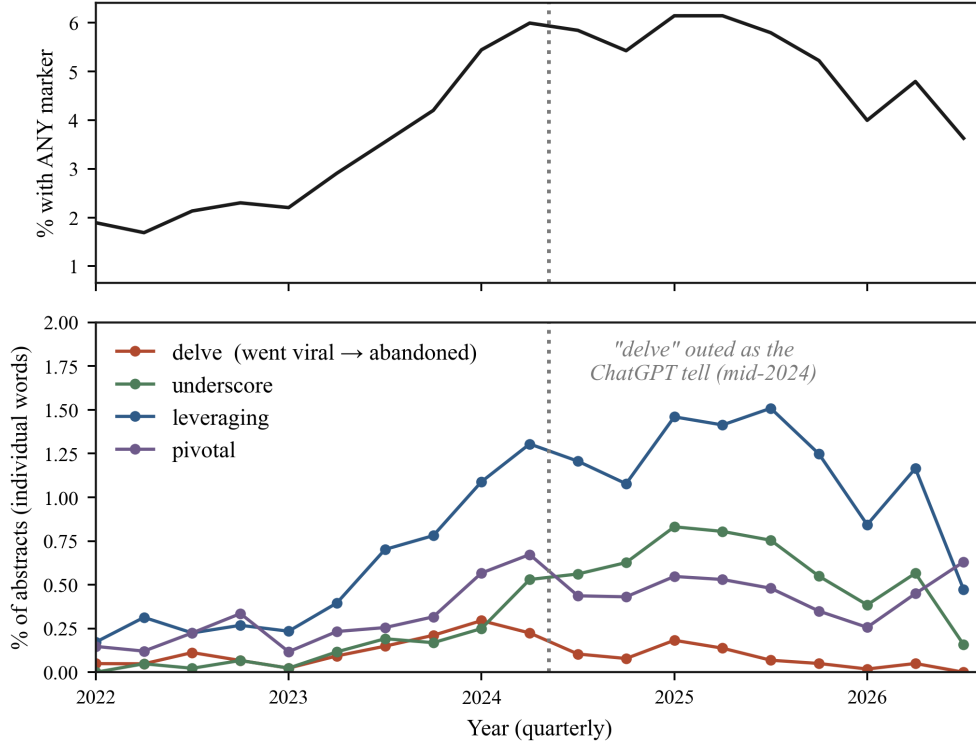
We queried the arXiv API for every paper with a primary or cross-listed astro-ph category, month by month, from January 2015 through July 2026, and kept the version-1 submission date, the primary category, and the abstract. After removing duplicate versions this gives 200,547 abstracts. The API returns the abstract of the latest posted version, so a paper revised after 2022 can carry recent language back to its original submission date. This smears the signal backward in time and biases the measured rise down; the flat pre-2022 marker rate in Section 3 shows the effect is small. The 2026 bin covers January through July. We tokenize each abstract into lowercase alphabetic words. A word’s “document frequency” in a year is the fraction of that year’s abstracts that contain the word at least once, which is the quantity used in earlier work.

For disclosure (Section 7) and for the by-country cut (Section 10) we use the NASA Astrophysics Data System, whose `ack` field searches the acknowledgments section and whose `full` field searches full text where available. We count `database:astronomy` records per year as the denominator.

## 3. THE MARKER-WORD RISE

We assembled a basket of marker words from earlier studies: `delve`, `underscore`, `intricate`, `pivotal`, `showcasing`, `meticulous`, and about thirty more. For a control we used neutral astronomy words such as `observed`, `measured`, and `galaxy`, which should not depend on writing tools. Figure 1 shows the two curves. The marker basket sat between 1.0% and 2.0% over 2015–2022, with a slow drift of 0.12 percentage points per year that we return to in Section 6.1, and rose to  $5.80 \pm 0.16\%$  by 2025. Against the pooled 2018–2021 baseline of 1.50%, the 2025 rate differs by  $z = 35$  in a two-proportion test. The mean frequency of the control set changed by  $+0.10$  percentage points between the same two periods, forty times less than the basket change, although individual control words drift by up to 2 percentage points for topical reasons (`obtained` fell 2.1 points; `redshift` rose 1.7). The rise is specific to the model-favored vocabulary and is not a general change in how abstracts are written.

The aggregate rise hides a moving target. Figure 2 tracks individual words by quarter. The word `delve` rose to a peak in early 2024 and then fell back toward its starting level once it became a widely known sign of model text,



**Figure 2.** Top: the fraction of abstracts with any marker word, by quarter, keeps rising. Bottom: single words. **delve** peaks in early 2024 and then declines, while other markers keep rising; the aggregate does not fall with it.

while less famous words such as **underscore** and **leveraging** kept rising. The total marker rate did not fall with **delve**; the load shifted to other words. Naming a marker removes it, but the underlying habit stays, which matches the cross-field result of Geng & Trotta (2025).

#### 4. LETTING THE DATA PICK THE WORDS

An imported word list invites the question of whether astronomy differs. We set the list aside and ranked every word by how much more common it became in 2024–2026 than in 2018–2021. The top of the ranking splits cleanly into two families (Figure 3). One family is the names of instruments and surveys that came online in these years (JWST’s NIRCam, MIRI, and NIRSpec; the DESI survey; XRISM), which is what a working method should find. The other family is the same style vocabulary reported in computer science and biomedicine, now recovered from astronomy on its own: **leveraging**, **pivotal**, **intricate**, **highlighting**. The imported list was not needed; the astronomy text points at the same words. Table 1 lists a representative set with their document frequency before and after ChatGPT.

The same split holds for two-word phrases. Figure 4 tracks a set of model-favored bigrams over time; **underscores the** and **highlighting the** rose by an order of magnitude, while a neutral control phrase (**wide range**) stayed flat. Ranking all bigrams by their increase again separates topic phrases (**jwst observations**, **the desi**) from style phrases (**offering a**, **results highlight**, **findings indicate**), as shown in panel (b) of Figure 3. Phrases carry a stronger signal than single words, and they are harder to remove by editing one word at a time.

#### 5. THE MARKER WORDS TRAVEL TOGETHER

If the marker words shared a common cause, they should occur together more often than chance. For a pair of words  $a$  and  $b$  we measured the lift

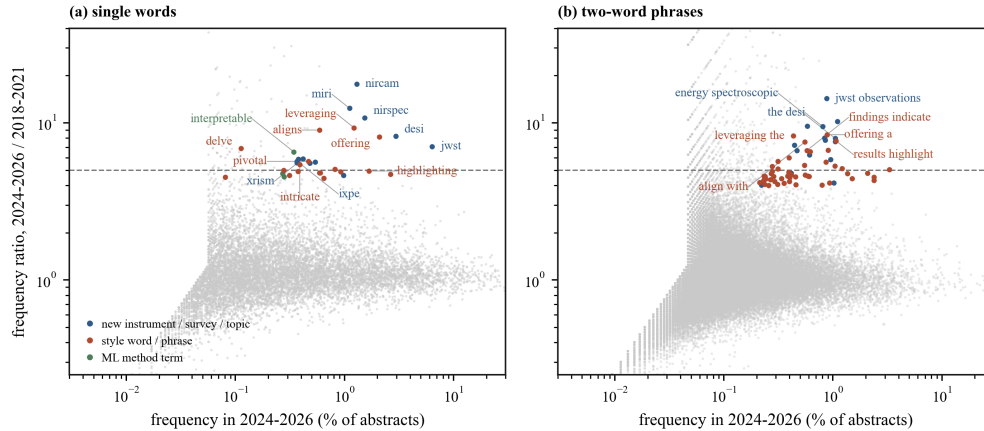
$$\text{lift}(a, b) = \frac{P(a \cap b)}{P(a)P(b)}, \quad (1)$$

where  $P(a)$  is the fraction of 2024–2026 abstracts that contain word  $a$  and  $P(a \cap b)$  the fraction that contain both. A lift of one means the two words are independent.

**Table 1.** Representative marker words: document frequency in the pre-ChatGPT baseline (2018–2021) and the recent window (2024–2026), and the ratio of the two.

| word                | 2018–21 (%) | 2024–26 (%) | ratio |
|---------------------|-------------|-------------|-------|
| <b>underscore</b>   | 0.03        | 0.58        | 23    |
| <b>showcasing</b>   | 0.01        | 0.12        | 10    |
| <b>leveraging</b>   | 0.13        | 1.22        | 9     |
| <b>offering</b>     | 0.26        | 2.08        | 8     |
| <b>delve</b>        | 0.02        | 0.11        | 7     |
| <b>pivotal</b>      | 0.08        | 0.47        | 6     |
| <b>highlighting</b> | 0.34        | 1.68        | 5     |
| <b>intricate</b>    | 0.07        | 0.39        | 5     |

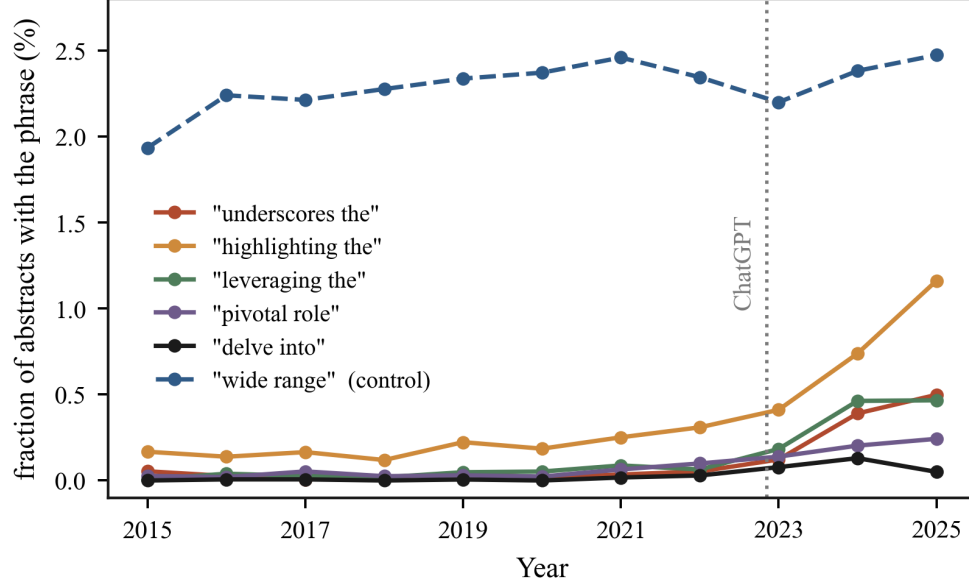
NOTE—Document frequency is the fraction of abstracts that contain the word at least once. The full table of every word that clears a count floor is in the public data release.



**Figure 3.** Frequency ratio between 2024–2026 and 2018–2021 against recent frequency for (a) every word and (b) every two-word phrase in the corpus (grey points). The tail above the dashed line at ratio 5 separates into new instruments, surveys, and topics (blue), style words and phrases (red), and technical machine-learning terms (green); selected items are labeled. Axes are logarithmic.

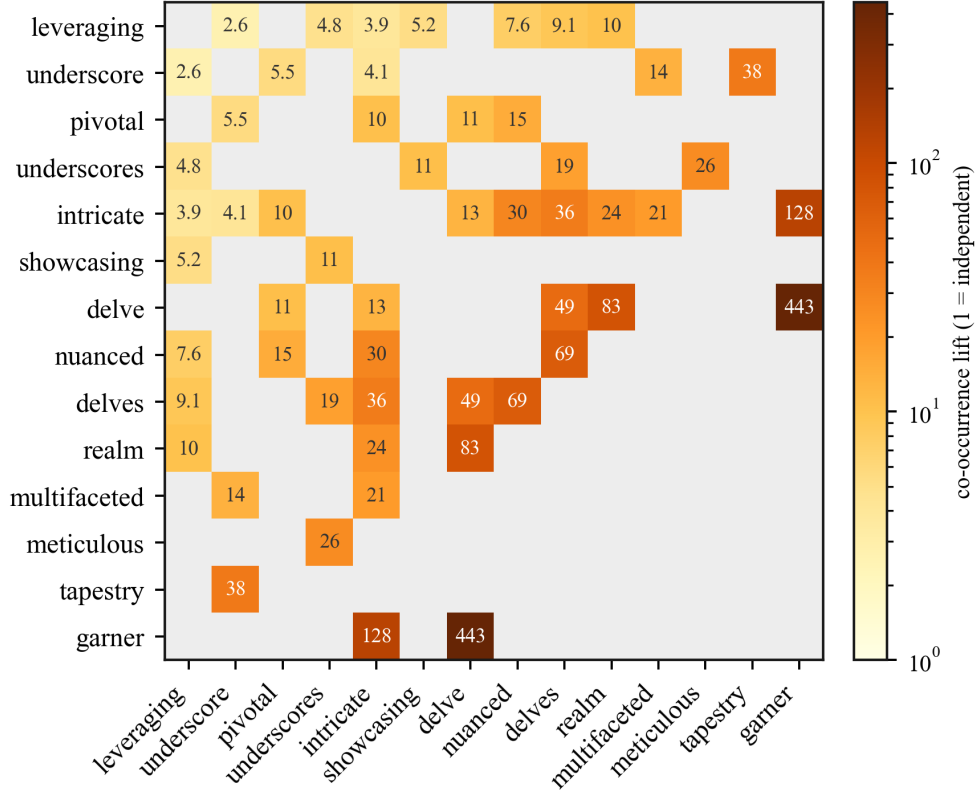
Because several markers are rare, many pairs have small joint counts, and a large lift can rest on a handful of abstracts. We therefore test each pair against a Poisson null: with 53,192 recent abstracts, the expected joint count under independence is  $\mu = NP(a)P(b)$ , and we ask for the probability of the observed count or more. Of the 91 pairs, 27 exceed independence at  $p < 0.01$ , where 0.9 false positives would be expected; Figure 5 shows only these significant cells. Among pairs with at least five joint occurrences the lift runs from 3 to 36 (**delves**+**intricate**: lift 36 from 5 joint abstracts against an expectation of 0.14,  $p = 4 \times 10^{-7}$ ; **intricate**+**pivotal**: lift 10 from 10 against 0.97,  $p = 8 \times 10^{-8}$ ). The very large lifts among the rarest words rest on one to three abstracts each and should be read as order of magnitude only, although even a single joint occurrence is significant when the expectation is  $2 \times 10^{-3}$ . The mean pairwise lift is 14, and 11 when restricted to pairs with at least five joint occurrences.

Independent word drift would not produce this pattern. A model that edits or drafts a whole abstract, and adds several of its preferred words at once, does. Part of the clustering could also be topical, since marker use is not uniform



**Figure 4.** Fraction of abstracts containing selected two-word phrases, by year. Model-favored phrases rise after 2022; a control phrase (**wide range**) stays flat.

across subfields (Section 10), but the strongest pairs join words with no shared subject matter. The clustering supports reading the marker signal as one effect rather than a set of unrelated trends.



**Figure 5.** Co-occurrence lift for pairs of marker words in 2024–2026 abstracts. Colored cells are the 27 of 91 pairs whose joint count exceeds the independence expectation at  $p < 0.01$  (Poisson test); grey cells are consistent with independence. Lifts above  $\sim 40$  rest on one to three joint abstracts and are order-of-magnitude values.

## 6. A LOWER BOUND ON THE TOUCHED FRACTION

To turn the signal into a fraction we count the share of abstracts that contain at least one word from a small basket of strong, low-baseline markers, and subtract the pre-2022 rate. Let  $f_t$  be the fraction of abstracts in period  $t$  that contain at least one basket word and let  $f_0$  be the same fraction in the 2018–2021 baseline. Our estimate is

$$\alpha_{\text{lb}} = f_t - f_0. \quad (2)$$

This is a lower bound for two reasons: a model-edited abstract need not keep any word from a short list, so  $f_t$  misses some model-touched abstracts; and by subtracting  $f_0$  we discard any human writer who uses a basket word for ordinary reasons. Both push the estimate down. For 2025 we find  $f_t = 5.8\%$  and  $f_0 = 1.5\%$ , so  $\alpha_{\text{lb}} = 4.3\%$ , and the true value is higher.

The basket is chosen for low baseline and high specificity rather than for size; broader baskets give larger numbers but add words with mixed meanings. Equation 2 is the  $q = 1$  limit of a two-population model: if a fraction  $\alpha$  of abstracts are model-edited and carry a basket word with probability  $q$ , while the rest carry one at the baseline rate  $f_0$ , then  $f_t = (1 - \alpha)f_0 + \alpha q$  and  $\alpha = (f_t - f_0)/(q - f_0)$ . Since  $q \leq 1$ ,  $\alpha \geq f_t - f_0$ ; any realistic  $q < 1$  raises  $\alpha$  above our bound.

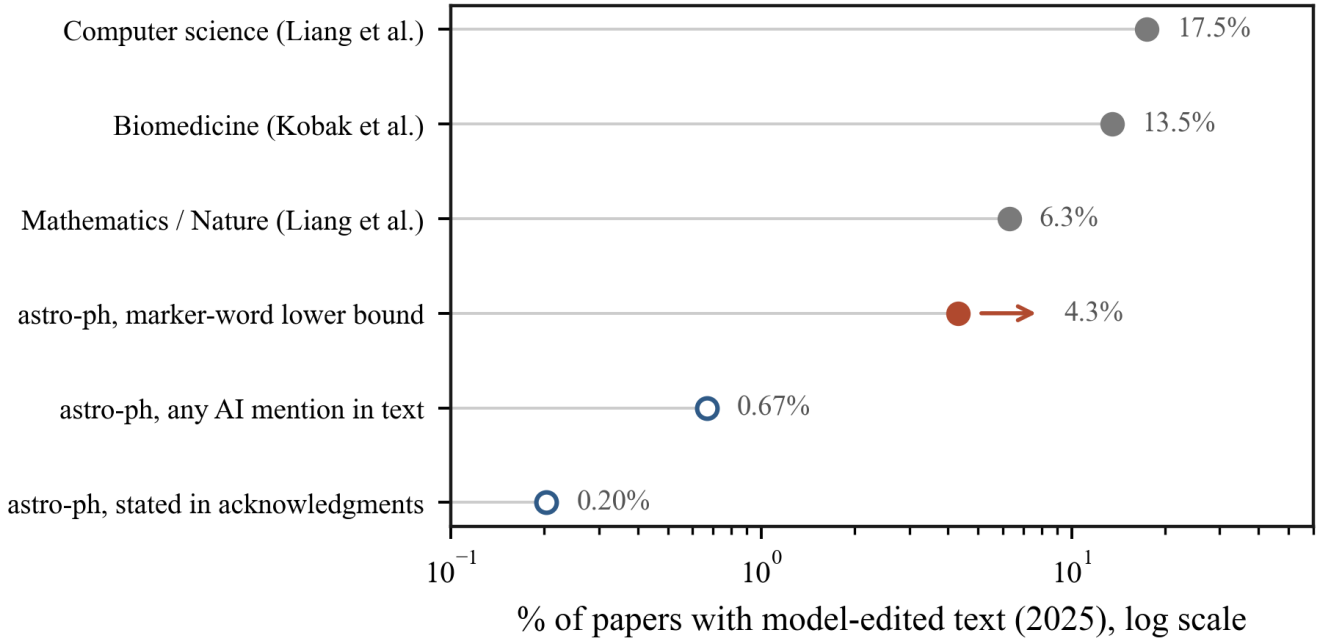
### 6.1. Robustness

Table 2 collects the variations we tried, computed for the pooled 2024–2026 window; for 2025 alone the default choices give  $\alpha_{\text{lb}} = 4.30 \pm 0.16\%$ , where the error is the binomial error on the two proportions. Moving the four-year baseline window earlier raises the pooled bound by at most 0.36 percentage points, because the basket rate drifted up slowly even before ChatGPT (0.12 points per year over 2015–2021). Treating that drift as a secular trend and extrapolating it linearly raises the effective 2025 baseline to 2.2% and lowers the 2025 bound from 4.3% to 3.6%, which

**Table 2.** Robustness of the lower bound  $\alpha_{lb}$ , pooled 2024–2026 window.

| variant                                               | $\alpha_{lb}$ (%) |
|-------------------------------------------------------|-------------------|
| default (baseline 2018–2021)                          | 3.9               |
| baseline 2017–2020                                    | 4.1               |
| baseline 2016–2019                                    | 4.2               |
| baseline 2015–2018                                    | 4.3               |
| worst leave-one-word-out ( <b>leveraging</b> removed) | 3.0               |
| control words, same estimator                         | 0.1               |

NOTE—For 2025 alone the default gives  $4.30 \pm 0.16\%$ , and a linear extrapolation of the pre-2022 drift in place of the flat baseline gives 3.6%.

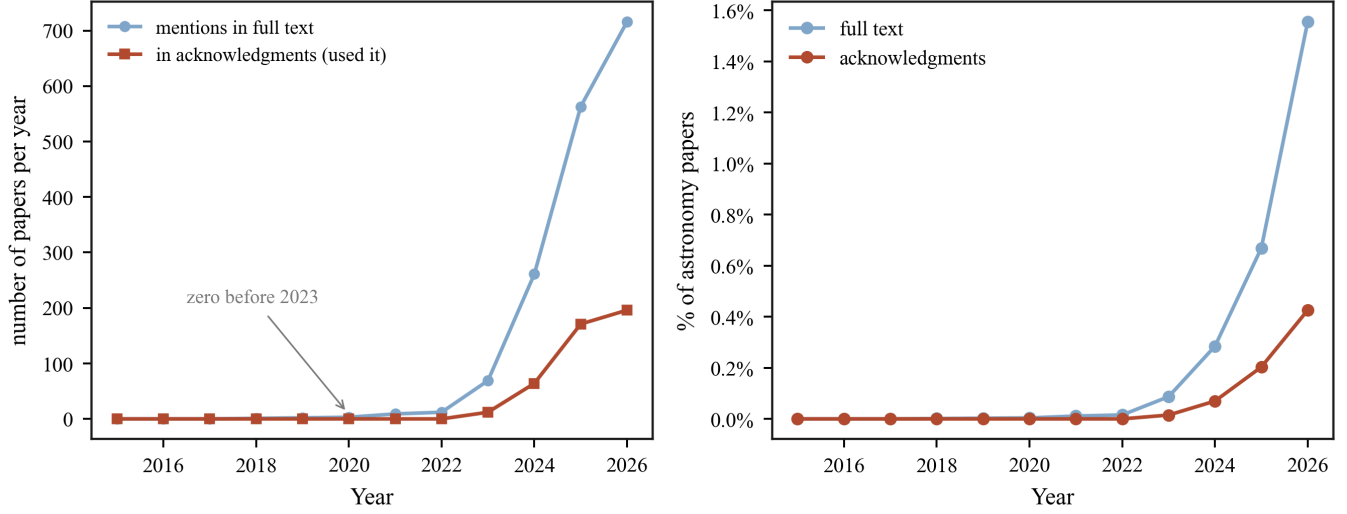


**Figure 6.** The model-touched fraction of 2025 papers across fields and detection routes, on a logarithmic axis. Grey points are published estimates for other fields (Liang et al. 2024; Kobak et al. 2025); the red point is our marker-word estimate for astro-ph, with the arrow marking it as a lower bound; open circles are the fractions of astronomy papers that mention or state model use. The gap between the red point and the open circles separates use from disclosure.

we consider the most conservative reading. Removing any single word from the basket changes the pooled bound by at most 1.0 point; the largest single carrier is **leveraging**, without which it is 3.0%. The control-word set run through the same estimator gives +0.10 points. Under every variant the bound stays between 3.0% and 4.3%.

Figure 6 places this number against published estimates for other fields. Astronomy sits below computer science and biomedicine and in the same range as physics and mathematics. Because our number is a strict lower bound while the others use fuller estimators, the honest reading is that astronomy is in the same range as the rest of science, at a few percent to low double digits.

## 7. USING A MODEL AND SAYING SO



**Figure 7.** Left: number of astronomy papers per year that mention a model in full text (blue) or state model use in the acknowledgments (red); both counts are zero before 2023. Right: the same as a share of all astronomy papers, which stays below 1%.

Authors who use a model can state so in the acknowledgments. We searched astronomy papers in the NASA Astrophysics Data System for clear terms (ChatGPT, GPT-4, “large language model”, GitHub Copilot), and left out terms that are ambiguous in astronomy, such as “Gemini” (the observatory) and “Claude” (a personal name). The count is exactly zero before 2023, which is a good check that the search is clean, and it rises every year after (Figure 7). As a share of all astronomy papers it stays below 1%: about 0.20% of 2025 papers state model use in the acknowledgments and about 0.67% mention a model anywhere in the text. Full-text coverage in the database is incomplete, so these fractions are themselves lower limits, but the missing coverage would have to be large and heavily skewed to close the factor-of-twenty gap to the word signal. Set against the 4% floor from the word counting, model use is common and stated use is rare.

## 8. A LADDER OF WAYS TO CATCH IT

arXiv posts the LaTeX source of most papers. A model’s boilerplate is sometimes pasted into that source and left in a comment that never reaches the PDF, such as “as an AI language model”. We downloaded the source of 3,950 papers, sampled with a fixed stride within each year, and searched for such text. In the 2,085 papers from 2024–2026 we find 3 hits, a rate of 0.14% (Wilson 95% interval 0.05–0.42%), and two of the three are papers that study models as a research topic rather than writing slips. The 1,244 pre-2023 papers give zero hits (95% upper limit 0.31%), so the search itself is clean.

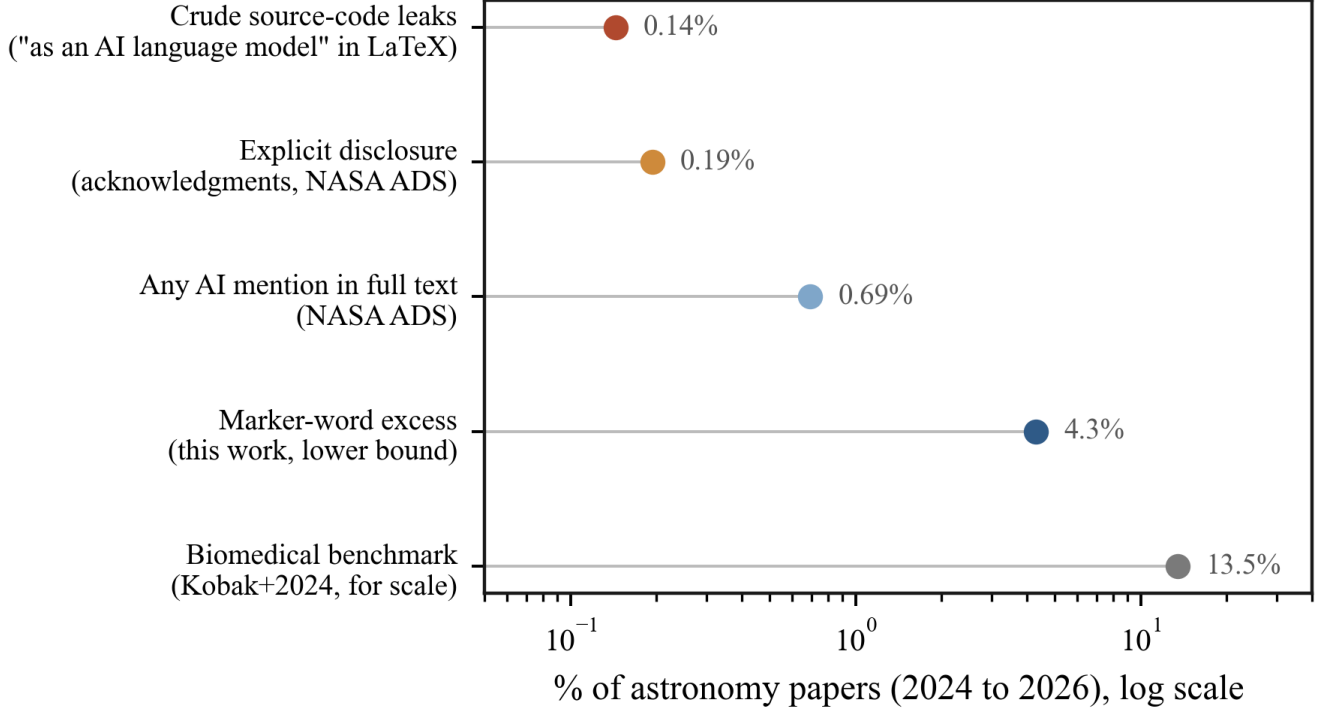
Placing the four lenses side by side (Figure 8) shows a range of two orders of magnitude on the same corpus. Crude source leaks (0.14%) and stated disclosure (0.19%) find the least; the word signal (4.3%, a lower bound) finds far more. A reader who trusted only the crude checks would conclude that astronomy has almost no model use, which is wrong. The word counting is the only method here that sees most of the use.

## 9. CITATION INTEGRITY

Models sometimes produce citations to papers that do not exist, and the rate of such citations is rising in biomedicine: from about one paper in 2828 in 2023 to one in 458 in 2025 and one in 277 in early 2026 (Topaz et al. 2026). Astronomy has a defense: nearly every author builds reference lists from the NASA Astrophysics Data System, so the identifiers are real by construction. We tested this on the LaTeX source of 2,532 papers from 2022–2026, drawn with a fixed stride within each year and weighted toward 2024–2026.

We extracted every cited arXiv identifier and digital object identifier (DOI). The 22,547 cited new-style arXiv identifiers (16,850 unique) all resolve; none points to a paper that does not exist. Of the 56,706 unique cited DOIs we checked a random sample of 2,500 against Crossref, which returned 134 misses, and we inspected every one. Most are real records in registries that Crossref does not index: 87 arXiv DataCite identifiers, 22 Zenodo records, and 16 data-

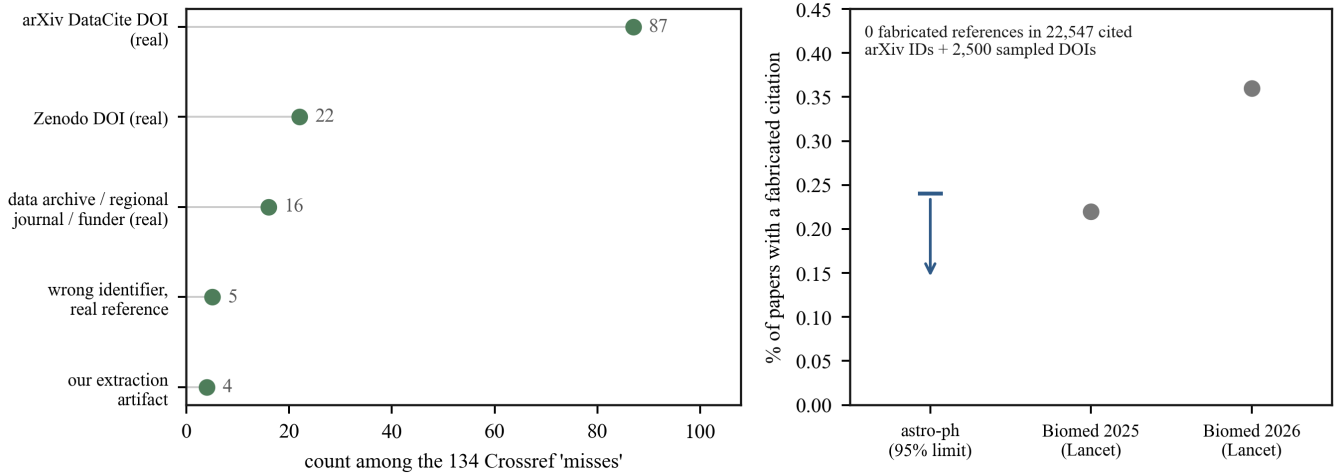




**Figure 8.** Four ways of looking for model use in the same 2024–2026 astronomy papers, on a log scale. Source leaks and stated disclosure find the least; the word signal finds most. The biomedical estimate (Kobak et al. 2025) is shown for scale.

archive, regional-journal, and funder identifiers, of which four we confirmed directly through the global handle resolver. Four more are artifacts of our own extraction. The remaining five are wrong identifiers attached to real references: two DOIs in a single bibliography whose suffixes do not exist while every other field (authors, title, journal, volume, page, arXiv number) matches a real paper, two identifier schemes that are not registered DOIs (an ACM proceedings prefix and an ISBN written as a DOI), and one journal citation-template placeholder. We find no fabricated references, in the sense of a reference to a work that does not exist.

The null result carries statistical weight. Restricting to the 1,250 papers with at least one checkable arXiv identifier, zero flagged papers gives a 95% upper limit of 0.24% on the per-paper rate. At the early-2026 biomedical rate of one paper in 277 we would expect 4.5 flagged papers here, so that rate is excluded at 95% confidence ( $p = 0.011$ ); the 2025 biomedical rate of one in 458 predicts 2.7 and is disfavored but not excluded ( $p = 0.065$ ). Per cited identifier the limit is  $1.3 \times 10^{-4}$ . Two caveats bound the claim. Our check tests whether a cited work exists, not whether it supports the claim it is attached to, and the wrong-suffix DOIs above show that identifier corruption does occur even when the reference is real. And the test covers papers that cite through arXiv identifiers, which skews toward recent, arXiv-native bibliographies (Figure 9).



**Figure 9.** Left: classification, after manual inspection, of the 134 DOIs that a Crossref check could not resolve. None is a fabricated reference: most are real records in registries Crossref does not index, five are wrong identifiers attached to real references, and four are artifacts of our extraction. Right: the fabricated-citation rate; the astronomy value is the 95% upper limit from zero detections (arrow), against the biomedical audit of [Topaz et al. \(2026\)](#).

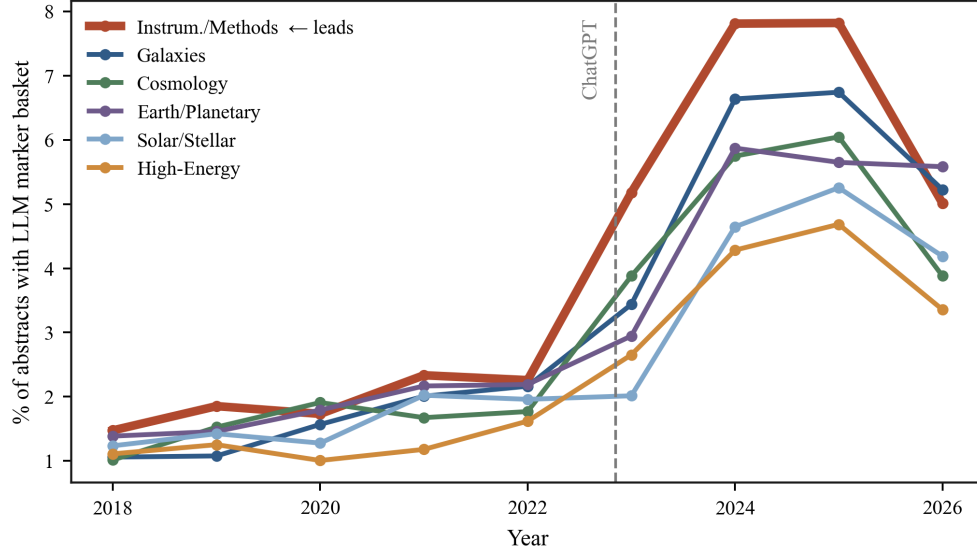
## 10. WHO, AND WHERE

The marker rate is not uniform. By subfield (Figure 10), the Instrumentation and Methods category, which is closest to software and machine learning, has the highest rate throughout, and high-energy astrophysics the lowest. The “computer science first” pattern seen across science also holds inside astronomy.

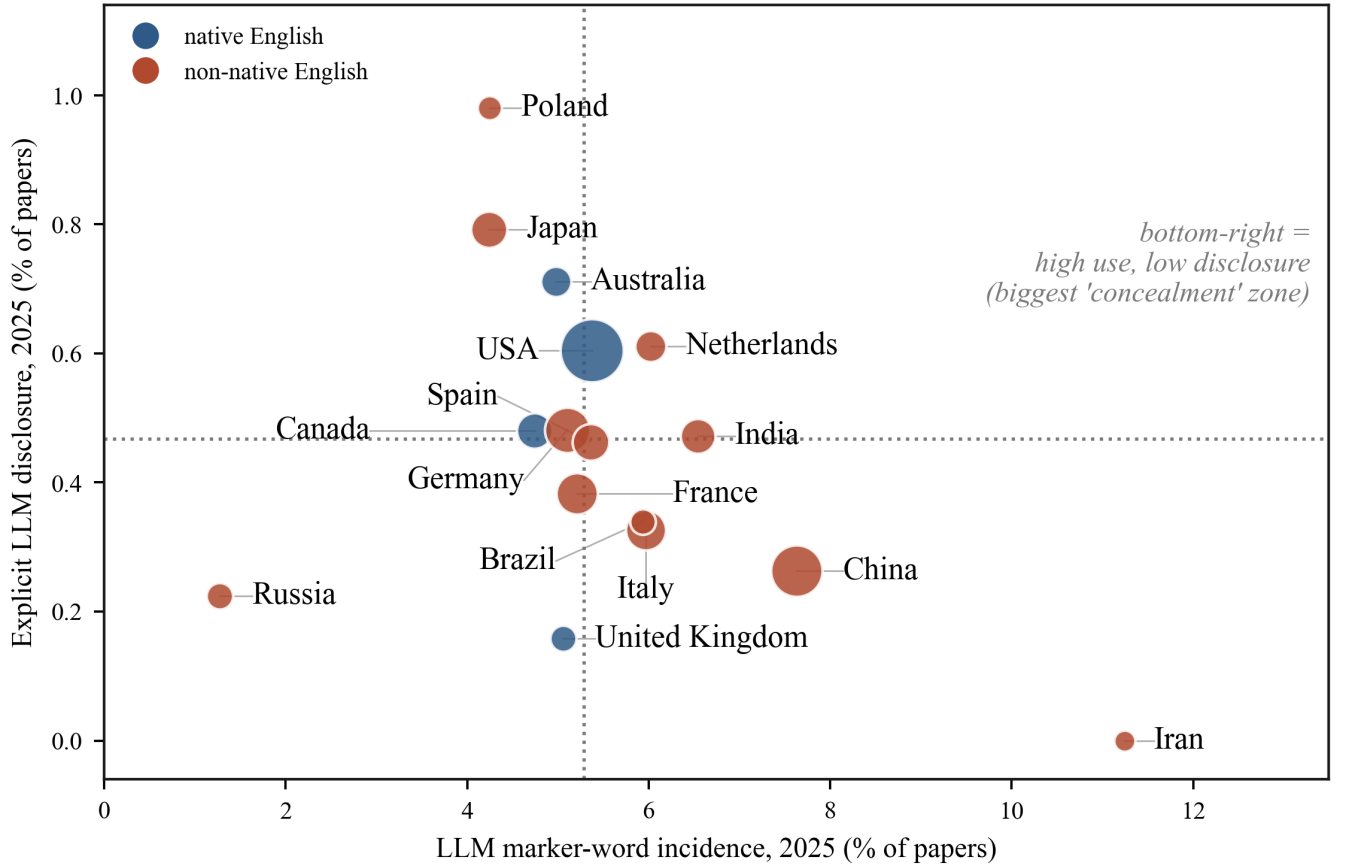
By author country (Figure 11), obtained from the affiliation field, papers from countries where English is not a first language tend toward higher marker rates, with Iran (11.2%, Wilson 95% interval 8.6–14.6%,  $n = 418$ ), China (7.6%,  $n = 8,377$ ), and India (6.5%,  $n = 3,179$ ) at the high end and Japan (4.2%) and Russia (1.3%) lower. This is consistent with a reading of these tools as a writing aid for non-native English speakers, and it is a reason to treat a marker word as evidence of editing rather than of anything worse. Papers from China combine a high marker rate (7.6%) with one of the lowest disclosure rates (0.26%), the widest gap in the sample. The affiliation search counts every country on a paper, so large collaborations appear more than once, and the small-sample countries are noisy; these numbers are first-order.

## 11. LOUDER, NOT MORE CAREFUL

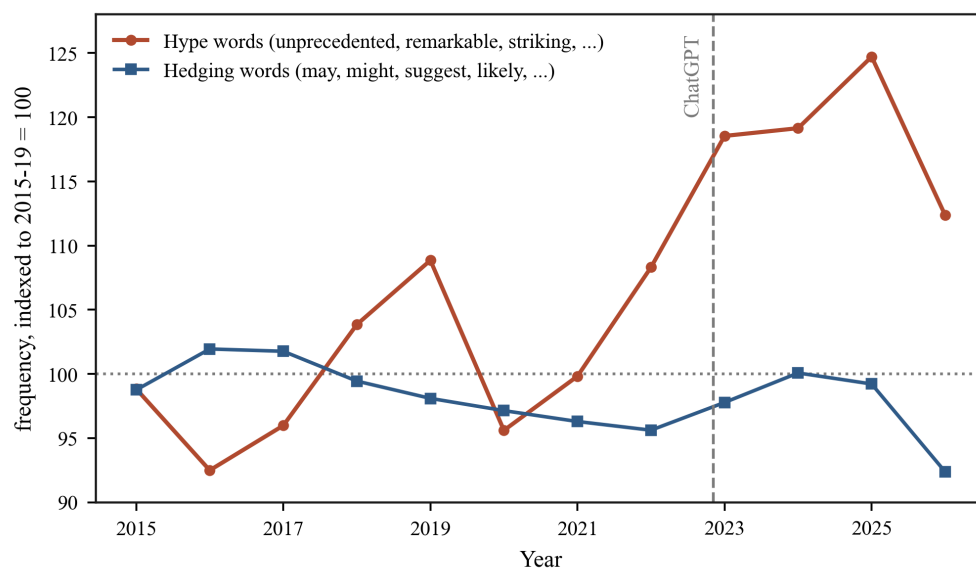
Word counts can measure tone as well as presence. We tracked promotional words (**unprecedented**, **remarkable**, **striking**) and hedging words (**may**, **might**, **suggest**) that mark caution. The promotional set rose by about 19% over the period and follows the marker rate closely; the hedging set did not move (Figure 12). Abstracts got louder without getting more careful. This is a correlation, and competition for attention pushes promotion up on its own, but the flat caution alongside rising volume is worth recording.



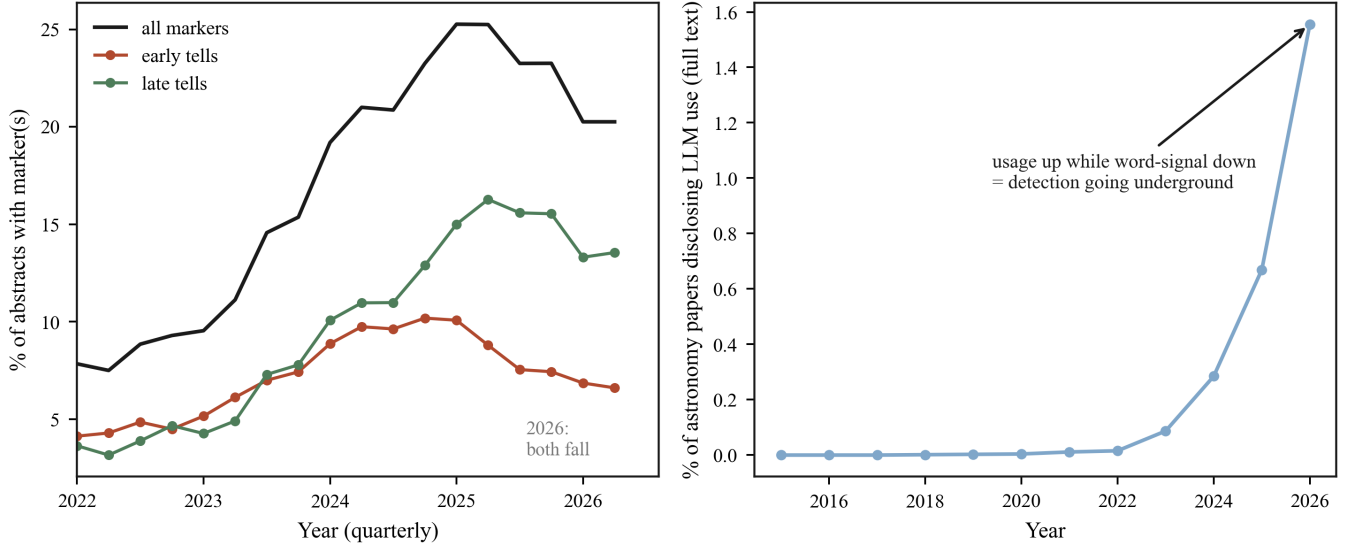
**Figure 10.** Marker rate by primary astro-ph subfield. Instrumentation and Methods leads; high-energy astrophysics trails.



**Figure 11.** Marker rate against disclosure rate by author country in 2025. Blue points mark countries where English is a first language, and bubble size is the number of papers. Wilson 95% intervals on the marker rates are quoted in the text; the smallest sample shown is Iran with  $n = 418$ . The affiliation search counts every country on a paper, so large collaborations enter more than once.



**Figure 12.** Promotional and hedging word frequency, indexed to the 2015–2019 mean. Promotional words rise with model use; hedging words stay flat.



**Figure 13.** Left: the marker words weaken across the board in 2026, both the early tells that peaked first (*delve*, *intricate*, *pivotal*) and the later ones (*underscore*, *leveraging*). Right: over the same months, stated model use keeps rising. Use is not falling; the word signal is.

## 12. THE SIGNAL IS WEAKENING

The marker rate dips a little in 2026. This is not a drop in model use. Figure 13 places the marker rate next to the disclosure count over the same months: the marker words soften across the board while stated model use keeps rising, already past the full 2025 count in half a year. What weakens is the word signal, not the use. Newer models write with fewer of the known words, and authors edit the famous ones out. Any word-based number is therefore a lower bound, and the bound will slide down even as use grows.

## 13. DISCUSSION

The measurements above give a consistent picture. A few percent of recent astro-ph abstracts, at least, carry model-edited language; the true fraction is higher and is in the range measured for other physical sciences. Stated use is rare, below 1%, so the gap between use and disclosure is large. The marker words rise together and cluster within abstracts, with 27 of 91 pairs above independence at  $p < 0.01$ , which points to a single cause rather than unrelated drift. Crude “gotcha” checks find almost nothing, so the population statistics are the only reliable lens. Astronomy so far avoids the fabricated-citation problem seen elsewhere, at a level we can bound below the early-2026 biomedical rate, most likely because of its shared reference database.

Several limits apply. First, a marker word measures model-edited language, not model-generated science; an abstract with *delve* in it is not fraudulent. Second, for many authors, and in particular for those writing in a second language, these tools lower a real barrier, so a marker word is better read as a sign of editing than of misconduct. Third, the word method has a short shelf life: as Section 12 shows, the known words fade as they become known, so a fixed list undercounts by more each year. Fourth, we read abstracts, not full papers, and abstracts are the most edited part of a paper. Finally, three uses that are easy to run together should be kept apart: a model as a copy editor, a model that drafts content, and a paper produced whole. The word counting mostly sees the first.

These results also point to method. Word lists will keep losing power, so the more durable signals are the ones tied to content rather than style: the co-occurrence structure of Section 5, which is harder to remove by editing single words, and integrity checks such as the citation audit of Section 9, which astronomy can run more cleanly than most fields because of the NASA Astrophysics Data System.

The gap between use and disclosure has a bearing on refereeing. If a few percent of abstracts are model-edited while under one percent say so, then a referee or an editor cannot assume that a clean, fluent text was written without help, and a screening tool that flags marker words will misjudge honest authors far more often than it catches misuse, because the base rate of misuse is low and the base rate of editing is not. The same tool applied to referee reports would

face the same problem. A disclosure norm, stated once at submission and kept simple, would remove the guesswork at low cost, and would separate the ordinary use of a model as an editor from the uses that raise real concern.

There is also a feedback loop to watch. As a growing share of the published corpus is model-edited, the next generation of models, and any astronomy-specific models trained on astro-ph text, will learn from text that already carries the style. Word-frequency screening will then measure the training data as much as the writing, and its output will drift. The signals that do not depend on a fixed word list, such as the co-occurrence structure and the citation checks, are the ones that will still mean what they say.

## 14. CONCLUSIONS

We measured the footprint of large language models in the astronomy literature from 200,547 astro-ph abstracts. The main results are:

1. The fraction of abstracts with a marker word rose from about 1.5% before 2022 to about 5.8% in 2025, while neutral control words stayed flat.
2. A conservative lower bound puts the model-touched fraction of 2025 abstracts near 4%; the true value is higher and is in the range measured for physics and mathematics.
3. Without an imported list, the data recover the same style words and phrases next to the genuinely new instrument and survey names, and the marker words occur together far more often than chance.
4. Stated disclosure stays below 1%, and crude source-code checks find near-zero pasted text; the word signal sees roughly two orders of magnitude more than either.
5. We find no fabricated citations, in contrast with a rising rate in biomedicine; the astronomy reference habit appears to protect the citation record.
6. The word signal weakens in 2026 even as disclosure rises, so word counts are a lower bound that will get harder to measure.

Whether astronomy should agree on a disclosure norm now, before a single case forces one, is a question the field can still answer on its own terms.

## APPENDIX

### A. THE MARKER BASKET

Table 3 lists the full 38-word basket used for the lower bound, with document frequencies in the baseline and recent windows. The control set is **observed**, **measured**, **obtained**, **presented**, **galaxy**, **stellar**, **sample**, **temperature**, **redshift**, and **spectra**.

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*Software:* NumPy, pandas, Matplotlib, PyArrow

## REFERENCES

- |                                                                       |                                             |
|-----------------------------------------------------------------------|---------------------------------------------|
| Astarita, S., Kruk, S., Reerink, J., & Gómez, P. 2024, arXiv e-prints | Geng, M., & Trotta, R. 2025, arXiv e-prints |
| Fouesneau, M., et al. 2024, arXiv e-prints                            | Gray, A. 2025, arXiv e-prints               |

**Table 3.** The marker basket: document frequency (per cent of abstracts) in 2018–2021 and 2024–2026.

| word         | 2018–21 | 2024–26 | ratio |
|--------------|---------|---------|-------|
| leveraging   | 0.131   | 1.224   | 9     |
| underscore   | 0.025   | 0.585   | 23    |
| pivotal      | 0.082   | 0.468   | 6     |
| underscores  | 0.030   | 0.449   | 15    |
| intricate    | 0.072   | 0.389   | 5     |
| encompassing | 0.133   | 0.352   | 3     |
| elucidate    | 0.181   | 0.340   | 2     |
| showcase     | 0.124   | 0.305   | 2     |
| underscoring | 0.010   | 0.278   | 27    |
| plethora     | 0.137   | 0.132   | 1     |
| unraveling   | 0.046   | 0.120   | 3     |
| showcasing   | 0.012   | 0.118   | 10    |
| unravel      | 0.094   | 0.115   | 1     |
| delve        | 0.016   | 0.113   | 7     |
| showcases    | 0.025   | 0.107   | 4     |
| elucidating  | 0.042   | 0.094   | 2     |
| nuanced      | 0.018   | 0.081   | 5     |
| seamlessly   | 0.034   | 0.073   | 2     |
| noteworthy   | 0.061   | 0.071   | 1     |
| delves       | 0.001   | 0.068   | 45    |
| seamless     | 0.024   | 0.068   | 3     |
| realm        | 0.073   | 0.060   | 1     |
| multifaceted | 0.013   | 0.047   | 3     |
| myriad       | 0.051   | 0.043   | 1     |
| garnered     | 0.012   | 0.041   | 3     |
| unravelling  | 0.015   | 0.041   | 3     |
| meticulous   | 0.009   | 0.032   | 4     |
| meticulously | 0.009   | 0.032   | 4     |
| intricacies  | 0.013   | 0.028   | 2     |
| showcased    | 0.003   | 0.026   | 9     |
| tapestry     | 0.001   | 0.017   | 11    |
| boasts       | 0.007   | 0.015   | 2     |
| delving      | 0.003   | 0.015   | 5     |
| realms       | 0.010   | 0.006   | 1     |
| commendable  | 0       | 0.004   | ...   |
| garner       | 0       | 0.004   | ...   |
| testament    | 0.001   | 0.004   | 3     |
| garners      | 0       | 0.002   | ...   |

Kobak, D., González-Márquez, R., Horvát, E.-Á., & Lause, J. 2025, *Science Advances*, 11  
Kurtz, M. J., Eichhorn, G., Accomazzi, A., et al. 2000, *A&AS*, 143, 41

Liang, W., Zhang, Y., Wu, Z., et al. 2024, arXiv e-prints  
Topaz, C. M., et al. 2026, *The Lancet*, 407,  
doi: [10.1016/S0140-6736\(26\)00603-3](https://doi.org/10.1016/S0140-6736(26)00603-3)